

Synergistic Speed Control Strategy for PMDC Motor

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Abstract—This paper presents the synthesis of a linear-quadratic-regulator (LQR) augmented with a nonlinear-integral controller to optimize the speed regulation of a permanent magnet DC (PMDC) motor. The LQR utilizes the full state-feedback and minimizes the quadratic performance index to deliver optimal speed control decisions. However, the LQR lacks the ability to eliminate the steady-state error and damp the oscillations occurring during transients. For this purpose, the LQR is combined with an integral controller. In order to synergize the controller combination, the integral gain is dynamically varied via smooth hyperbolic secant function of error-dynamics in motor speed; such that, the contribution rendered by the integral-action increases as the response converges towards the reference speed and vice-versa. The results of real-time experiments are presented and analyzed to validate the effectiveness of the proposed controller.

Keywords—PMDC motor; LQR; integral controller; hyperbolic secant function.

I. INTRODUCTION

The permanent magnet direct current (PMDC) motors convert electrical energy to rotational mechanical energy. They are easily controllable, cost-effective, and offer variable load-handling capabilities [1]. Owing to these features, they are widely used in automobiles, drilling machines, conveyors, disk-drives, and robots etc [2]. A lot of research has been done to synthesize optimal closed-loop speed controllers and DC motor speed drives [3-8]. Among these controllers, the linear-quadratic-regulators (LQR) are highly preferred [9]. They use the state-space model of the dynamical system along with the full-state feedback and minimize an energy-like performance index to generate optimal control effort. The LQR is effective in improving the transition time and error-convergence rate of the system [10]. However, due to inevitable modeling-errors, it lacks the ability to nullify the steady-state errors and the oscillations (or overshoots) incurred in the response during transients or random perturbations. This issue can be resolved, by augmenting the existing LQR controller with an additional integral actor [11]. Hence, in this research, an additional integral controller (acting upon the error in angular velocity) is combined with the LQR based DC motor speed controller. The integral-gain of the controller is adaptively modulated using a smooth hyperbolic secant function of error (in angular-velocity) to optimize only the steady-state response and provide favorable damping effect [12]. The bounded nonlinear gain function is designed such that it offers negligible integral action during transient conditions in order to speed up the

response time. However, as the response converges towards the reference speed, the integral gain increases in order to eliminate unnecessary overshoots, oscillations and steady state-errors. The proposed LQR with Nonlinear-Integral (LQNI) controller acts as a robust-optimal speed regulator.

II. EXPERIMENTAL SETUP

The details of the experimental platform are discussed in the following sub-sections.

A. Hardware setup

The control system is implemented on the DC Motor Control Trainer (DCMCT) setup by Quanser and is tested via 'hardware-in-the-loop' experiments [13]. It is shown in Fig. 1. The platform comprises of a PMDC motor with shaft encoder, tachometer, and current-sensing resistor (0.1 Ω). These sensors measure the angular velocity and the armature current of the motor. The motor shaft is coupled with an inertial load disc. The hardware setup is integrated with a LabVIEW based graphical user interface [14, 15]. This application serves to graphically visualize the variations in the system's state(s) and generate appropriate control commands to accurately track the reference speed. The NI ELVIS II board is used as a communication-relay to serially transmit the control commands and acquire the digital sensor readings at 9600 bps [16]. The control commands are fed to the motor via a linear pulse-width-modulated (PWM) power amplifier in order to control the speed and direction of motor rotation [17]. The zero-order hold method is used to discretize the system. The sampling time and PWM switching frequency used in this research is 2.0 ms and 10.0 kHz, respectively.



Fig. 1. QNET DC Motor Control Trainer

B. Mathematical Model

The torque and back-electromotive force of the motor are directly proportional to the armature current ' i ' and angular velocity of motor rotation ' ω_r ', respectively. Their expressions are given by (1) and (2), respectively.

$$T = K_b i \quad (1)$$

$$E_b = K_t \omega_r \quad (2)$$

The values of K_b and K_t are equal to each other. The differential equations representing the mechanical and electrical characteristics of the motor are given by (3) and (4), respectively [18]. The transfer function is given in (5).

$$J \frac{d\omega_r}{dt} + b\omega_r = K_b \quad (3)$$

$$L \frac{di}{dt} + Ri = V_m - K\omega_r \quad (4)$$

$$\frac{\omega_r(s)}{V_m(s)} = \frac{K_b}{(Js + b)(Ls + R) + K_b^2} \quad (5)$$

The state-space model of a linear dynamical system is generally given by (6) and (7).

$$\frac{dx}{dt} = Ax + Bu \quad (6)$$

$$y = Cx + Du \quad (7)$$

where, x is the state-vector, y is the output-vector, u is the control input signal, A is the system matrix, B is the input matrix, C is the output matrix, and D is the feed-forward matrix. The state-vector of motor is given by (8).

$$x = [\omega_r \quad i]^T \quad (8)$$

The matrices A , B , C , and D in the motor's state-space model are identified in (9). The motor voltage (V_m) acts as the control input (u) of the motor.

$$A = \begin{bmatrix} -\frac{b}{J} & \frac{K_b}{J} \\ -\frac{K_b}{L} & -\frac{R}{L} \end{bmatrix}, B = \begin{bmatrix} 0 \\ \frac{1}{L} \end{bmatrix}, C = [1 \quad 0], D = 0 \quad (9)$$

The motor parameters are identified in Table I.

III. PRELIMINARIES

This section provides theoretical background of the PI as well as LQR based DC motor speed controllers.

TABLE I. MOTOR PARAMETERS

Parameters	Symbol	Values
Armature Resistance	R	3.3 Ω
Armature Inductance	L	0.047 H
Maximum Torque	T_{max}	0.14 Nm
Torque Constant	K_b	0.028 Nm/A
Back-EMF Constant	K_t	0.028 V/(rad/s)
Motor Inertia	J	9.64 $\times 10^{-6}$ kgm ²
Rated Voltage	V_s	± 24.0 V
Rated Current	I_s	5.0 A
Inertial Load Disc Mass	m	0.033 kg
Inertial Load Disc Radius	r	0.0242 m

A. Proportional-Integral Controller

The proportional-integral (PI) controller is widely favored in industries. It is preferred due to its simple structure, as shown in Fig. 2, and effective controlling ability [19, 20]. The derivative controller is usually neglected because it tends to significantly amplify high frequency noise signals as well. The integral controller is proportional to the magnitude and duration of the error in the actual angular speed (ω_r). Furthermore, it effectively and robustly nullifies the small constant offsets and errors occurring in the steady-state response. It ensures accurate tracking of reference speed signal (ω_d) by providing the motor with additional torque to compensate the perturbations.

The PI controller is linear combination of the error, shown in (10), and integral of errors. The weightages of the aforementioned error-dynamics are denoted as the proportional (k_p) and integral (k_i) gains, respectively. The control law is thus given by (11).

$$e(t) = \omega_d(t) - \omega_r(t) \quad (10)$$

$$u(t) = k_p e(t) + k_i \int_0^T e(\tau) d\tau \quad (11)$$

The PI gains must be properly tuned so they may yield good control performance. In this research, these gains are manually tuned via trial-and-error technique to yield optimal performance. The cost function, given in (12), is used to attain the PI gains with highest fitness.

$$J_{cost} = (OS)^2 + (t_r)^2 + (t_s)^2 \int_0^T e(\tau)^2 d\tau \quad (12)$$

where, OS is the overshoot, t_r is the rise-time, and t_s refers to the time taken by the response to settle within $\pm 1.0\%$ of reference speed. The optimal set of PI gains, yielding the minimum value of cost function, is given in Table II.

In practice, the control input to the motor tends to reach the actuator limits due to unprecedented dynamical variations or occurrence of large input errors. Consequently, the feedback loop terminates and the system continues to operate in open-loop. In order to overcome this abnormality, the control input is restricted between an upper and lower bound, as shown in (13). This saturation keeps the control input to the motor from abnormally increasing or decreasing.

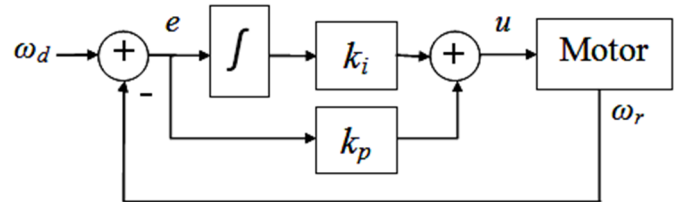


Fig. 2. Proportional-Integral controller

TABLE II. CONTROLLER PARAMETERS

Parameter	Value	Fitness ($1/J_{cost}$)
k_p	0.75 V/(rad/s)	0.026
k_i	1.85 V/rad	0.035

$$u' = \begin{cases} u, & \text{for } V_{\min} < u < V_{\max} \\ V_{\max}, & \text{for } u \geq V_{\max} \\ V_{\min}, & \text{for } u \leq V_{\min} \end{cases} \quad (13)$$

where, $V_{\min}$ and $V_{\max}$ are the minimum and maximum allowable voltage input to the motor terminals. In this research, the voltage input is bounded between ± 10.0 V. However, using the aforementioned saturation technique, the integrator output keeps accumulating if the error in the controlled state continues to remain non-zero during the saturation period [21]. This phenomenon is denoted as “integral wind-up”. It commonly occurs in the systems that have integral control action and actuator saturation. It renders severe degradation and delays in the system’s response. In order to alleviate the problems caused by the integral wind-up, the back-calculation and tracking technique is incorporated with the integral control scheme [22]. The weighted difference between the actual and the saturated control output is fed back to re-adjust the integral control term in order to compensate the effects of integral wind-up. The diagrammatic illustration of PI controller with the proposed anti-windup compensator (AWC) is shown in Fig. 3. The gain (k_r) is experimentally tuned at 0.235, by minimizing the cost function in (12).

B. Linear Quadratic Regulator

The LQR is a model-based controller that uses the state-feedback of a linear dynamical system, shown in (6), to minimize a quadratic cost function in order to generate optimal control decisions [23]. The LQR structure is shown in Fig. 4. The LQR system is described via the Algebraic Riccati Equation, as shown in (14).

$$A^T P + PA - PBR^{-1}B^T P + Q = 0 \quad (14)$$

For a symmetric positive definite matrix, P , the state-feedback gain vector, K , is given by (15)

$$K = R^{-1}B^T P \quad (15)$$

The gain vector, K , helps to relocate the open-loop poles of the system to synthesize an optimal controller. The quadratic cost function is expressed in (16).

$$J = \int_0^T x^T Q x + u^T R u \, dt \quad (16)$$

where, Q and R are the state and control penalty matrices, respectively. They are chosen such that; $Q = Q^T \geq 0$ and $R = R^T > 0$. In this research, these matrices are heuristically tuned. The minimization criterion of the cost function in (12) is used to optimize the matrices. The matrices are expressed in (17).

$$Q = \begin{bmatrix} 0.12 & 0 \\ 0 & 0.09 \end{bmatrix}, \quad R = 0.11 \quad (17)$$

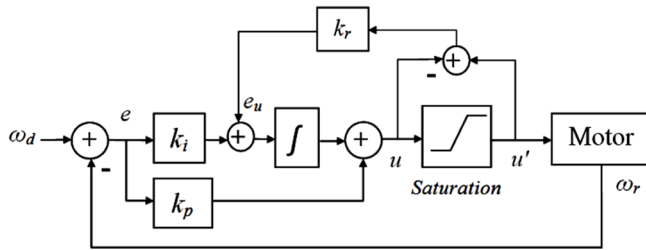


Fig. 3. PI controller with anti-windup compensation

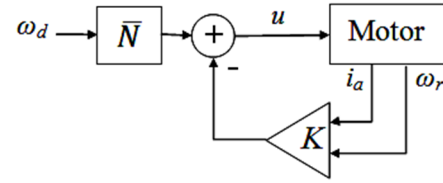


Fig. 4. Linear Quadratic Regulator

Owing to the quadratic nature of the cost function, the control signal, u , is proportional to the squared of variations in the states. Thus, if the state-variations are large; the minimization is faster. Consequently, the LQR system quickly converges towards an optimal gain vector (K) and delivers optimal control commands, as shown in (18).

$$u(t) = -Kx(t) \quad (18)$$

This phenomenon significantly improves the transient response and disturbance rejection capability of the controller. The reference input is fed and subtracted from the LQR control output using the constant gain, $\bar{N}$, having a value of 0.032 V/(rad/s). Pertaining to the system description given in the previous section, the optimal state-feedback gain vector, yielded by the LQR system, is given in (19).

$$K = [1.245 \quad 0.385] \quad (19)$$

IV. CONTROLLER SYNTHESIS

The LQNI controller is synthesized by augmenting the conventional LQR controller with an integral control term that uses the error in motor’s angular-speed. The PI controller designed in the previous section could also be used; however, the control effort delivered by the LQR is directly proportional to the variations in the ω_r . Hence, employing an additional proportional controller that is serving the same purpose is overkill. Consequently, only an integral-control term is merged with the LQR controller [24].

Since the specialty of the integral controller is to eliminate steady-state errors, inhibit overshoots and damp the oscillations; therefore, it is mostly needed when the response is in close proximity of the reference speed. An optimally tuned fixed gain integral controller would serve all the aforementioned objectives. However, it would also simultaneously and inevitably hamper the control performance of LQR. Owing to the integral action, the overall transition time and disturbance rejection capability of the system would be slightly degraded as compared to the performance originally exhibited by the LQR alone. Hence, in order to synergize the controller performance, the integral gain is dynamically updated so that its contribution, in optimizing the response, may gradually increase as the response draws closer to steady state (or as the response converges to reference speed after recovering from a disturbance or transients). The fuzzy inference systems or artificial neural networks are normally used to self-tune the controller gains (and other parameters) [25, 26]. The fuzzy inference systems rely upon heuristically developed logical rule-bases which may not be able to cater all possible uncertainties incurring in the complex dynamical systems. Similarly, the artificial neural networks require large sets of training data for their synthesis. The

smooth nonlinear functions offer a tangible self-tuning capability as well. They have a simple structure and are easy to compute. Several nonlinear functions are presented in the literature [27]. In this research, a bounded hyperbolic secant function is used for the adaptive tuning of integral gain [28]. The hyperbolic secant function is expressed in (20).

$$k_i(e) = 0.05 + 1.91 \operatorname{sech}(\alpha \times e(t)) \quad (20)$$

where, α is the variation-rate of the nonlinear integral gain functions, and $\hat{e}$ is the normalized-error given by (21).

$$\hat{e}(t) = \left| \frac{e(t)}{\omega_d(t)} \right| \quad (21)$$

The value of α is heuristically tuned at 18.5 by using the minimization criterion of cost function given in (12). The waveform of the nonlinear integral gain function is shown in Fig. 5. When the error is large, the integral gain is kept smaller so that only the LQR may act to quickly rise towards the reference speed. As the response converges towards the reference, the integral gain is gradually increased in order to increase the motor damping and eliminate unnecessary overshoots and oscillations. As the response settles, the integral controller continues to nullify the small steady state fluctuations. The proposed LQNI control law is given in (22) and is diagrammatically shown in Fig. 6.

$$u(t) = -Kx(t) + k_i(e) \int_0^T e(\tau) d\tau \quad (22)$$

The LQNI controller is retrofitted with the AWC as well, in order to suppress the effects of actuator saturation and integral wind-up phenomenon.

V. TESTS AND RESULTS

The performance of the LQNI speed controller is analyzed and compared with the performances exhibited by the fixed-gain PI controller, the LQR, and the LQI controller (an LQR equipped with a fixed-gain Integral controller and AWC) using the following two testing-scenarios. The test-trials are conducted in real-time as ‘hardware-in-the-loop’ experiments [29]. The resulting responses are graphically visualized and recorded.

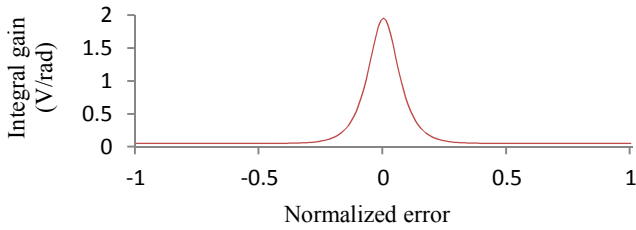


Fig. 5. Nonlinear integral gain function

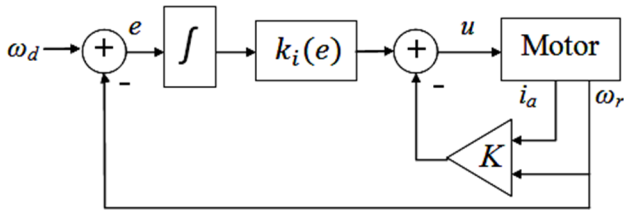


Fig. 6. Proposed LQNI controller

Test-case A: A step-input of 100 rad/s is applied to the motor. The resulting variations in ω_r for each controller, is shown in Fig. 7. The magnified steady-state response of each controller is presented in Fig. 8. The corresponding variations in the integral gain of LQNI controller are shown in Fig. 9. The summary of test-results is presented in Table III.

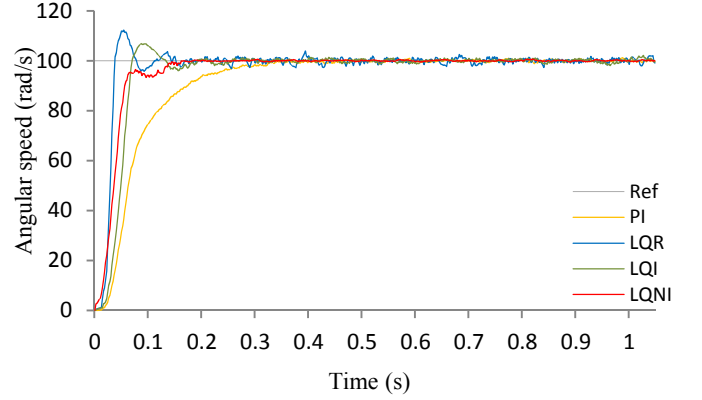


Fig. 7. Step-response of the controllers

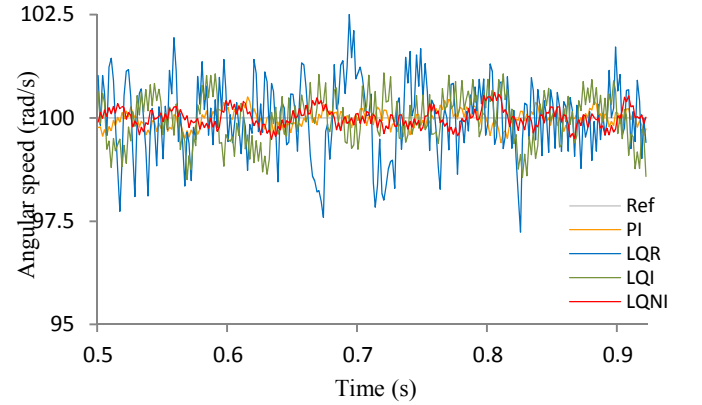


Fig. 8. Expanded view of the steady-state response

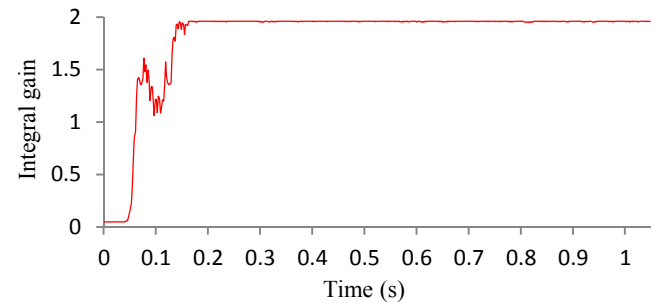


Fig. 9. Variation in integral gain under normal conditions

TABLE III. SUMMARY OF STEP-RESPONSES WITHOUT DISTURBANCE

Controller	t_r (s)	OS (%)	t_s (s)	E_{ss} (rad/s)
PI	0.23	0.00	0.43	0.28
LQR	0.06	12.24	0.13	1.10
LQI	0.11	6.85	0.24	0.69
LQNI	0.10	0.00	0.18	0.31

Test-case B: An artificially generated bounded impulsive disturbance signal is applied at the controller's output to perturb the steady-state step-response of the motor. The amplitude and duration of the disturbance signal is +1.0 V and 0.01 s, respectively. The abrupt variations occurring in ω_r are illustrated in Fig. 10. The corresponding variations in the integral gain of LQNI controller are shown in Fig. 11. In this case, the responses are analyzed by evaluating the maximum-absolute-error (MAE) incurred in the response upon being disturbed, and the time taken to recover and converge to $\pm 1.0\%$ of reference speed (t_{rec}) after being disturbed. The summary of the performance analysis is recorded in Table IV.

VI. CONCLUSION

This paper presents a speed regulation mechanism that synergistically combines the correctional capabilities offered by an optimal LQR with a nonlinear-integral controller. The augmentation of the state-feedback with a nonlinear integral controller significantly improves the transient response, steady-state response, and disturbance rejection capability of the system.

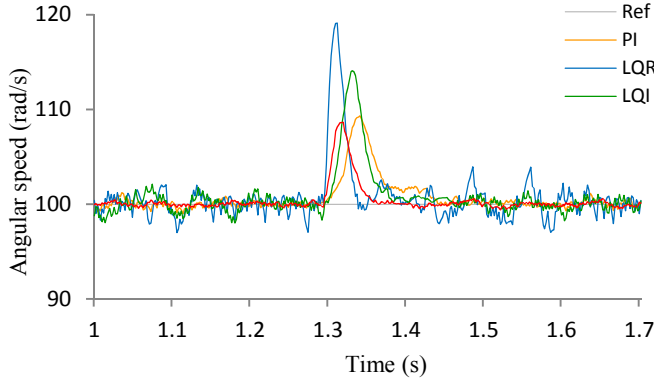


Fig. 10. Effect of impulsive disturbance

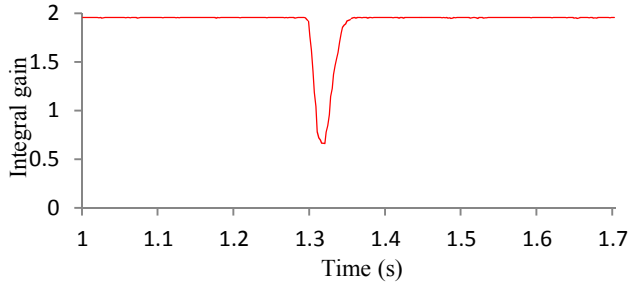


Fig. 11. Variation in integral under gain under disturbance

TABLE IV. SUMMARY OF STEADY-STATE RESPONSES WITH DISTURBANCE

Controller	t_{rec} (s)	MAE (rad/s)
PI	0.13	9.22
LQR	0.04	19.13
LQI	0.10	14.10
LQNI	0.05	8.75

The dynamic variation in the control contribution rendered by the nonlinear self-tuning of the integral gain enhances the accuracy of the speed control response. The LQNI control architecture beneficially utilizes the available control resources. Thus, it is simple and robust. The comparative assessment of LQNI controller with the rest of the controllers clearly manifests that the proposed controller yields a superior performance. In future, other nonlinear functions can be used instead of the hyperbolic secant function to dynamically update the integral gain. Moreover, the nonlinear gain function can be retrofitted with the information regarding the derivative-of-error, apart from the error itself, in order to optimize the dynamic adaptation of controller gains. Adaptive combination of self-tuning integral controllers with other model-based control techniques can also be investigated to enhance the motor's speed control performance.

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